Midterm Quiz

(Please answer all questions in English)

Some useful physical constants:

Electronic charge, $q = 1.6\times10^{-19}$ C
Boltzmann's constant $k = 1.38 \times 10^{-23}$ J/K
Permittivity of free space, $\varepsilon_0 = 8.854\times10^{-14}$ F/cm
Free electron mass, $m_0 = 9.1\times10^{-31}$ kg
Thermal voltage $kT/q = 26$ mV at 300K

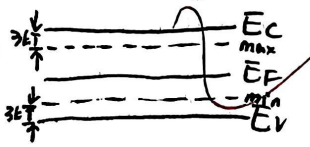
Properties of silicon (Si) at 300K:

Energy band gap, $E_G = 1.12$ eV
Intrinsic carrier concentration, $n_i = 10^{10}$ cm^{-3}
Permittivity, $\varepsilon_{Si} = 1.0\times10^{-12}$ F/cm
Effective DOS of conduction band, $N_C = 2.8\times10^{19}$ cm^{-3}
Effective DOS of valence band, $N_V = 1.04\times10^{19}$ cm^{-3}

Please answer the following questions:

(1) What is the meaning of non-degenerated doped semiconductor? Where is its range of Fermi level (E_F) located with respect to E_C and E_V? Plot schematically, if you like.

The non-degenerated doped semiconductor means that the doping is not very heavy.
The range of E_F is $E_V+3kT < E_F < E_C-3kT$. ⇒



(2) What is the dopant ionization energy?

The dopant ionization energy means the doping materals ionized its electrons.
Exciting the electrons to the vancuum energy (真空能级). ~~during this, the~~
The energy needed by this procession (过程) called dopant ionization energy.
(The energy that ionized the dopant's one single electrons to the vancum level).

(3) What is the fundamental cause of phonon scattering of carriers in a semiconductor? How is such scattering process influenced by temperature?

① Fundamental cause is that the mobile carriers have collision with the vibration crystal lattice, create phonons. whose energy is quanity. (量子化的) ~~晶格~~ vibration crystal energy
② The temperature increase, then made the collision ~~longer~~ more and ~~longer~~ more (更多碰撞). thus making the phonon scattering increase. (the velocity of the mobile carriers increase)

(4) Wide-bandgap semiconductors (such as SiC, GaN, etc.) are typically used in high-temperature applications. Based on what you learned in class, give one reason why this makes sense.

Because the wide-bandgap can also work well in high-temperature because its wide bandgap. If the Eg not that large. in high-temperature, due to the thermal velocity of electrons (电子热运动), the elections can easily get the energy larger than Eg. thus making many election are excited (激发). The device can't be work. So the wide-bandgap materials make sense in working in high-temperature.

(5) In a PN junction, the p-side is rich in what type of dopant (doner / acceptor)? In the depletion region, the n-side is occupied by ionized impurity atoms which are positively charged / negatively charged. (Please circle the correct answer in the underlined content.)

① P-side is rich in acceptor.
② n-side is occupied by ionized impurity atoms which are positively charged.

(6) In the n-side of a PN junction, what conditions must be met (in terms of carrier concentrations), so that "low-level injection" occurs?

low-level injection must let the excessively carriers $|\Delta p| = |\Delta n| << n_{no}$.

$|\Delta p|$ and $|\Delta n|$ must be very very less than the majority carriers's concentration (n_{no} in n-side)

(7) What is the cause of velocity saturation in semiconductor? How would it impact the performance of electronic device?

① The cause is when the velocity increase, the Kinetic energy (E_K, 动能) increase to a critical level, there will generate phonons which prevent the E_K increase. (phonon 分走 E_K 能量) This phenomenon makes velocity saturation.

② the velocity is releated to mobility and current. the velocity has limit means the mobility and Current have limit, which means the device made by the semiconductor have limited speed. to turn on/off the device. limit the work frequeacy and device speed.

(8) What is the difference between carrier direct recombination and carrier recombination through R-G center?

① The carrier direct recombination means that the electron-hole pair are directly recombination at one "place"! ② but the R-G center, due to. reflect or impurity or surface state, generate a E_T between the E_c and E_v, the the electrons can move from E_c to E_T, and then from E_T to E_v, indirectly recombination with holes.

(9) What is the thermal equilibrium condition in a semiconductor? List three key characteristics that a semiconductor behave at thermal equilibrium.

① no Electric field
② no $\vec{B}$ (磁场)
③ no light or 辐射
④ no Strain (no external force) (无外加应力)

At thermal equilibrium. ④ Net current is zero.
① Combination rate equals to recombination
② There's only one Fermi level E_F
③ $n_0 p_0 = n_i^2$; $n = n_i e^{(E_F - E_i)/kT}$; $p = n_i e^{(E_i - E_F)/kT}$

(10) Describe the fundamental cause of built-in potential in a PN junction.

When the p-type and n-type first touched to each other. The carrier concentration gradient will let electrons in n type move to p-type; holes in p-type move to n-type. And then, due to the movement of carriers. The n-type materials combine(结合) the holes, and have positive ionized atoms (正离子) left. Also, the p-type materials combine the electrons, and have negative ionized atoms (负离子) left. The ionized atoms generate a Electric field which prevent the concentration gradient to let carriers move.

(11) Explain the two mechanisms that a PN junction has capacitive effect. In a reverse-biased P^+N junction, how is the capacitance impacted by the doping concentration on each side of the junction?

① One mechanisms is depletion capactive effect. It was caused by the depletion width changed with the external Voltage (外界电压). The Voltage changed. the depletion width changed and also the depletion width in p-side and nside are changed, thus making the Charge changed. ⇒ have capactive effect. ② The other is diffusion Capactive effect. In formal biased, the charge density is $dQ = \tau_t \cdot dI$, relation with Current. the external Voltage Changed ⇒ the current change ⇒ Charge density change. Thus it also have capacitive effect.

(b). In a reverse-biased P^+N junction. the capacitience only have depletion Capacitience. The C equals to ⇒ $C = A\sqrt{\frac{q N_A \varepsilon_s}{2(V_{bi} + V_R)}}$. $\frac{1}{N} = \frac{1}{N_D} + \frac{1}{N_A}$. P^+N material means that $N_A >> N_D$. So the p side doping concentration are significant. (N_A). $N \approx N_A$ N_A increase ⇒ C_{dep} increase; N_A decrease, C_{dep} decrease. The n-side dopants are donor N_D. But the N_D influnce the C_{dep} very very little that we could not see the difference when change N_D. Although (尽管) are the same

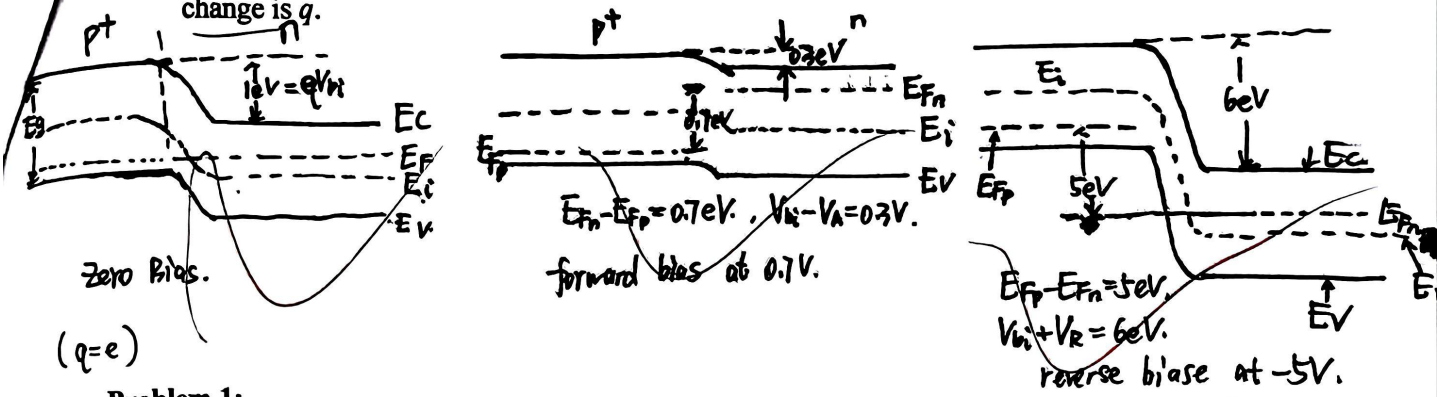
After thermal equilibrium, the field and concentration gradient (平衡了). Then, the field called built-in Electric field. The potential caused by the field called built-in potential

(12) What are the fundamental forces behind carrier drift and carrier diffusion processes in a semiconductor, respectively?

① Carrier drift ⇒ fundamental forces is Electric field.

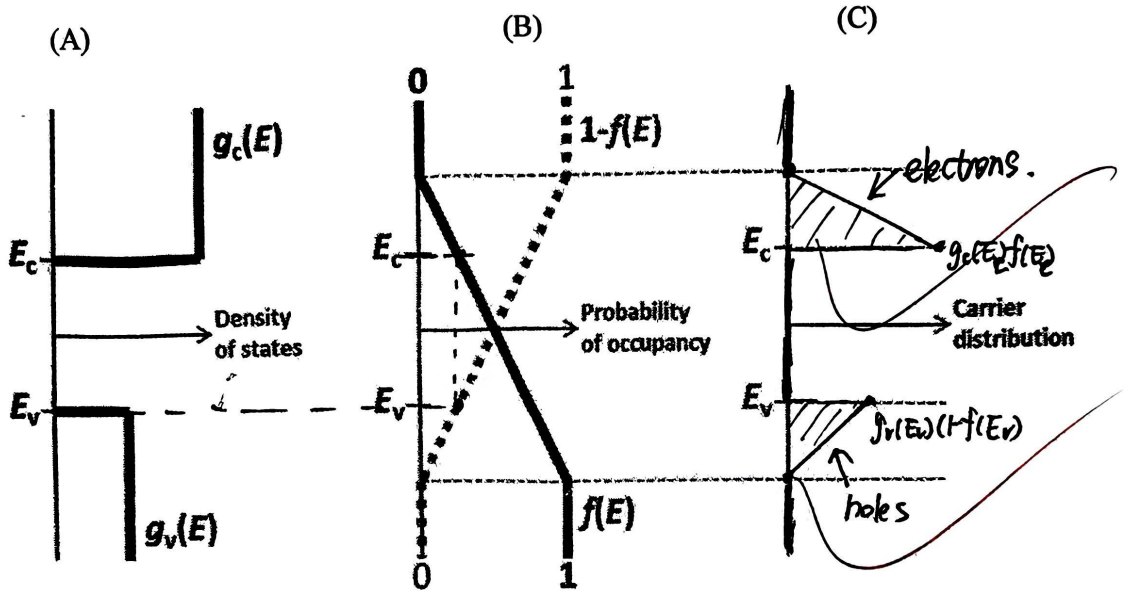
② Carrier diffusion ⇒ fundamental forces is ~~doping~~ Carrier concentration gradient (梯度) and the temperature gradient (Actually temperature impact concentration and then impact the diffusion)

(13) Schematically draw the energy band diagram of a P⁺N junction at zero bias, forward bias at 0.7V, and reverse bias at -5V, respectively. You need to show the relative positions of E_C, E_V, E_i, and E_F to include as much information as you can. Assuming the built-in voltage is 1V. The electronic change is q.



Problem 1:

A semiconductor material has the given functions of (A) density of states and (B) probability of occupancy, as shown below.



(a) On the plot (C) above, schematically draw the distributions of electron and hole concentrations as the functions of energy E within the conduction and valence bands, respectively.

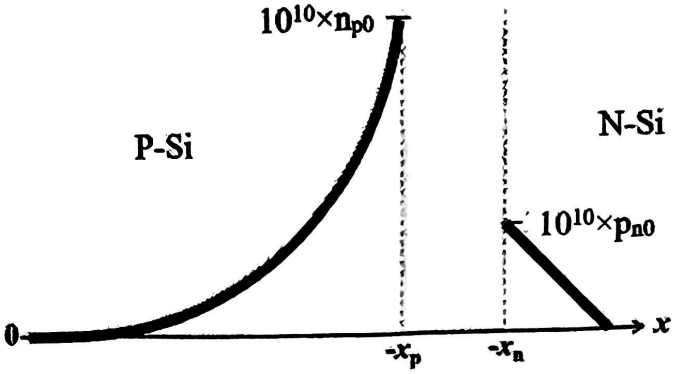
(b) Is this semiconductor material N-type or P-type? Why?

(b). N-type materials. because ~~the electron carrier distribution~~ the area of electron distribution with E(横坐标) is larger than ~~the those~~ that of holes. ⇒ means the concentration of electrons is larger than holes ⇒ $n > p$ ⇒ N-type materials

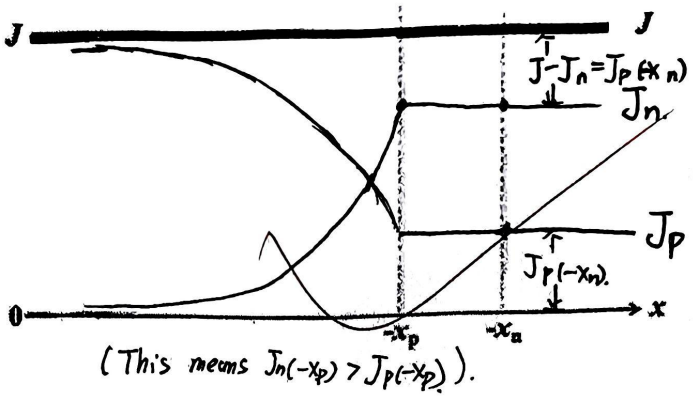
Problem 2:

The excess minority carrier concentrations within the quasi-neural regions of a silicon PN step junction maintained at T = 300K are shown below in (A):

(A)



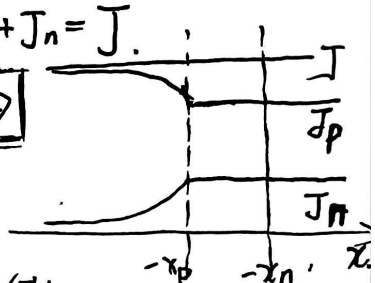
(B)



(This means $J_n(-x_p) > J_p(-x_p)$).

$J_p+J_n=J$.

or ⇒



(This means $J_n(-x_p) < J_p(-x_p)$).

(a) What is the value of the applied bias voltage, V_A? Is it forward-biased or reversed-biased? Why?
Note: $(kT/q)\cdot ln(10) = 0.06$ V

① $\Delta n = n_{p0}(e^{\frac{qV_A}{kT}} - 1) = 10^{10}\cdot n_{p0}$

$\Rightarrow e^{\frac{qV_A}{kT}} = 1+10^{10} \approx 10^{10}$

$\Rightarrow V_A = 10\cdot \frac{kT}{q} ln10 \approx 0.6V$ (>0)

② It's ~~reverse~~ forward-biased. Because if it's reverse-biased, V_A will below zero. which means $\Delta n \approx -n_{p0}$ is a constant. which is not right at this Graph.

(b) Schematically draw the hole and electron current density components (J_P and J_N) as function of location x on the plot (B) provided above.

$\begin{cases} J_p = -qD_p\cdot\frac{dp}{dx} = -qD_p\cdot\frac{d\Delta p}{dx} \\ J_n = qD_N\frac{dn}{dx} = q\cdot D_N\cdot\frac{d(\Delta n)}{dx} \end{cases}$ ⇒ from the equation we can get the plot (B) by the scope of plot (A). (斜率)

Short Diode $J = J_p + J_n$.

↓ for $J_n(x)$, $J_n = qD_N\frac{d(\Delta n)}{dx}$. ∴ $J_n(-x_p) = qD_N\frac{d(\Delta n)}{dx}\Big|_{x=-x_p} \approx 10^{10}\cdot q\frac{D_n}{L_n}n_{p0}$.

$J_n(-x_p) \approx 10^{10}\cdot q\cdot\frac{D_n}{W_n}\cdot n_{p0}$.

$J_p(-x_p) \approx 10^{10}\cdot q\frac{D_p}{L_p}\cdot p_{n0}$.

⇒ Can't Judge $J_n(-x_p)$? $J_p(-x_p)$ (无法判大小)

J_p 由于是缩短Diode. $\approx 10^{10}\cdot q\cdot\frac{D_p}{W_p}\cdot$

So in $x > -x_n$, the J_p and J_n are almost constant.